Einstein

Setup

In[ ]:=
```
<< mGRG`Einstein`
```

In[ ]:=
```
$PreRead = ReplaceAll[#,
    expr_String :> StringReplace[expr, "EP" -> "mGRG`Einstein`Private"]] &;
```

Component Operations

RiemannToWeyl and WeylToRiemann

In[ ]:=
```
RiemannToWeyl::usage
```

Out[ ]=
```
RiemannToWeyl[expr] converts
  the Riemann tensor in 'expr' to the Weyl tensor.
```

In[ ]:=
```
WeylToRiemann::usage
```

Out[ ]=
```
WeylToRiemann[expr] converts
  the Weyl tensor in 'expr' to the Riemann tensor.
```

Riemann 텐서 $R_{abc}{}^{d}$와 Weyl 텐서 $C_{abc}{}^{d}$를 서로 변환한다. 옵션으로 공변 도함수의 이름이 있다.

$$R_{abcd} = C_{abcd} + \text{(riemannSign)}\left(-\frac{1}{n-2}\left(g_{ac} R_{bd} - g_{ad} R_{bc} + g_{bd} R_{ac} - g_{bc} R_{ad}\right) + \frac{1}{(n-1)(n-2)} R\left(g_{ac} g_{bd} - g_{ad} g_{bc}\right)\right)$$

In[ ]:=
```
Format[GetDimension[Latin]] = n;
```

In[ ]:=
```
WeylCD[la, lb, lc, ld]
w2r = WeylToRiemann[%]
```

Out[ ]=

$$C_{abcd}$$

Out[ ]=

$$-\frac{g_{bd} R_{ac}}{-2+n} + \frac{g_{bc} R_{ad}}{-2+n} + \frac{g_{ad} R_{bc}}{-2+n} - \frac{g_{ac} R_{bd}}{-2+n} + R_{abcd} - \frac{g_{ad} g_{bc} R}{(-2+n)(-1+n)} + \frac{g_{ac} g_{bd} R}{(-2+n)(-1+n)}$$

In[ ]:=
```
w2r * Metricg[ub, ud] // Absorbg
```
Out[ ]=

$$R_{ac} + \frac{2\,R_{ac}}{-2+n} - \frac{\delta^{b}{}_{b}\,R_{ac}}{-2+n} - \frac{g_{ac}\,R}{-2+n} - \frac{g_{ac}\,R}{(-2+n)\,(-1+n)} + \frac{\delta^{b}{}_{b}\,g_{ac}\,R}{(-2+n)\,(-1+n)}$$

In[ ]:=
```
% /. Kdelta[ub, lb] → GetDimension[Latin] // Simplify
```
Out[ ]=

0

In[ ]:=
```
w2r * Metricg[ua, uc] // Absorbg
```
Out[ ]=

$$R_{bd} + \frac{2\,R_{bd}}{-2+n} - \frac{\delta^{a}{}_{a}\,R_{bd}}{-2+n} - \frac{g_{bd}\,R}{-2+n} - \frac{g_{bd}\,R}{(-2+n)\,(-1+n)} + \frac{\delta^{a}{}_{a}\,g_{bd}\,R}{(-2+n)\,(-1+n)}$$

In[ ]:=
```
% /. Kdelta[ua, la] → GetDimension[Latin] // Simplify
```
Out[ ]=

0

In[ ]:=
```
w2r
% // RiemannToWeyl
```
Out[ ]=

$$-\frac{g_{bd}\,R_{ac}}{-2+n} + \frac{g_{bc}\,R_{ad}}{-2+n} + \frac{g_{ad}\,R_{bc}}{-2+n} - \frac{g_{ac}\,R_{bd}}{-2+n} + R_{abcd} - \frac{g_{ad}\,g_{bc}\,R}{(-2+n)\,(-1+n)} + \frac{g_{ac}\,g_{bd}\,R}{(-2+n)\,(-1+n)}$$

Out[ ]=

$$C_{abcd}$$